

LECTURE 02 • WHEN WORDS GET RULES

Logic turns statements into objects

Propositions • connectives • truth tables • implication



LEARNING OUTCOMES

By the end, you can make conditionals behave

1 CLASSIFY

Recognize propositions, tautologies, and contradictions.

2 TRANSLATE

Turn everyday promises into logical formulas.

3 TEST

Use truth tables and counterexamples to compare conditionals.

4 REWRITE

Replace a conditional with an equivalent form without changing its truth.

Name the claim. Test every world. Trust the structure.

COLD OPEN

One samosa thief. One true statement.

Exactly one person took it, and exactly one statement is true. Who took the last samosa?

A I did not take the samosa.

B A took the samosa.

C I did not take the samosa.

C took it. Only A is telling the truth.

CASEWORK • TEST EACH CULPRIT

Only C leaves exactly one true statement

ASSUME A F • T • T 2 truths → reject	ASSUME B T • F • T 2 truths → reject	ASSUME C T • F • F 1 truth → survives
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Casework turns suspicion into certainty.

DEFINITION

A proposition makes one truth claim



declarative + fixed context + exactly one truth value

A proposition is true or false. Ignorance does not add a third option.

“Unknown to us” is not a third truth value. It is just Tuesday.

ONE DISTINCTION CHANGES EVERYTHING

Not knowing is different from not having a claim

UNKNOWN TO US

The 10¹⁰⁰th digit after the decimal point of π is 7.

Proposition: exactly one of T or F

OPEN SENTENCE

x stole the samosa.

Not a proposition until x is fixed

QUESTION

Who stole the samosa?

Asks for information; asserts nothing

A free variable leaves truth unfinished

$$x + 2 = 5$$

x has no value

Open sentence

$$3 + 2 = 5$$

→ **x replaced by 3**

Proposition: T

For every integer x , $x + 2 > x$

→ **x is bound by "every"**

Proposition: T

EXERCISE 1

Proposition or Not?

For each sentence below write whether it's a proposition.

01 Open the door. N Command. Repair: "The door is open."	02 x submitted before 5. N x is free. Repair: "Ayesha submitted before 5."
03 $2 + 2 = 5$. P False propositions still count.	04 The 100th digit after the decimal point of π is 7. P Unknown to us still means exactly one truth value.
05 Is the lab open? N Question. Repair: "The lab is open now."	06 Every student in this room submitted. P The group and moment are fixed.

THE PROPOSITION CHECK

Three checks prevent most category errors

1

Is it declarative?

Not a command or question

2

Is the context fixed?

No unnamed variable or comparison

3

Does exactly one truth value fit?

True or false—not neither, not both

False is allowed. Missing context is not.

MOVE 02 • COMPOSE

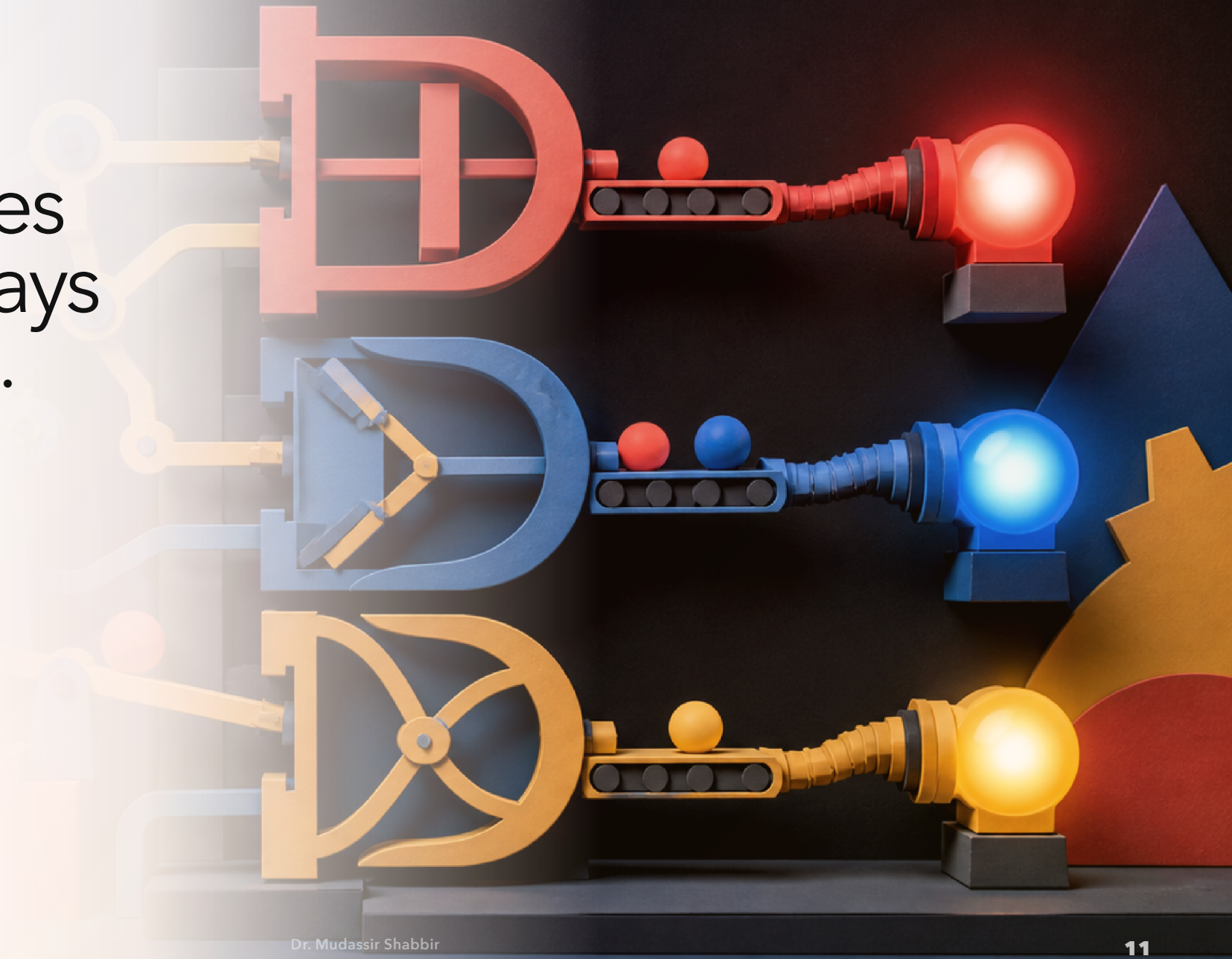
Statements to machines

T/F in. One new T/F out.

WHY FORMALIZE?

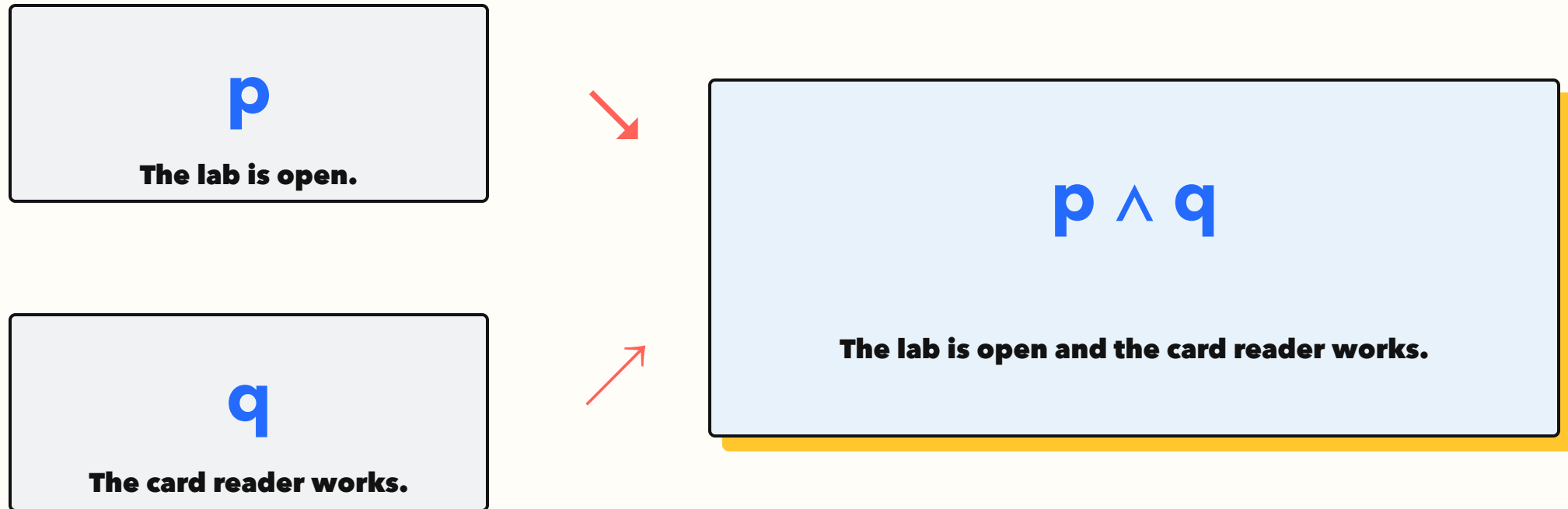
Formal logic scales where intuition frays

**Precise proofs • program verification •
digital circuits**



FROM SENTENCE TO SYMBOL

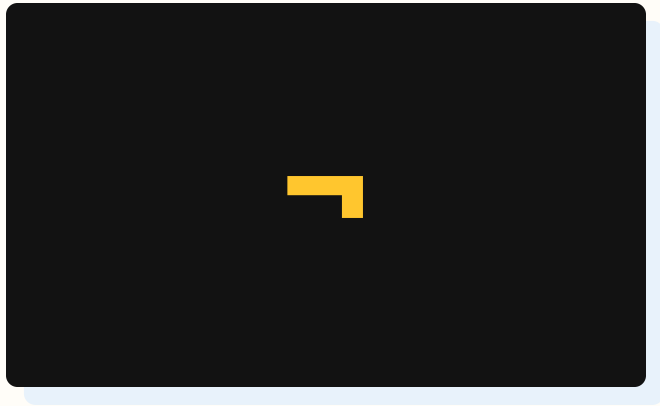
Name the atoms before you combine them



Two inputs feed one connective. The connective produces the output.

NEGATION • $\neg p$

NOT swaps T and F



T becomes F. F becomes T.

$\neg\neg p$ has the same truth value as p .

Double negation: logic's undo button actually works.

CONJUNCTION • $p \wedge q$

AND has one winning row


$$p \wedge q$$

True only when both inputs are true.

access = valid ticket $\wedge$ photo ID

DISJUNCTIONS

OR allows both; XOR rejects both

$p \vee q$

at least one

Both is allowed

$p \oplus q$

exactly one

Both is rejected

A waiter asks “tea or coffee?” Logic says yes. The waiter does not.

ORDER OF OPERATIONS

Operators evaluate in a fixed order

p = T

q = F

r = T

Parentheses override every default. Given **p = T, q = F, r = T**: name the main connective, then evaluate.

1 $()$
group

2 $\neg$
not

3 $\wedge$
and

4 $\vee$
or

$$\neg p \vee (q \wedge r)$$

MAIN $\vee$

LEFT $\neg p = \mathbf{F}$

RIGHT $q \wedge r = \mathbf{F}$

RESULT $\mathbf{F} \vee \mathbf{F} = \mathbf{F}$

EXERCISE 2

Four trays reveal four connective personalities

p = tray has tea **q** = tray has coffee

For each tray, predict **AND**, **OR**, **XOR**, and **neither**.

BOTH		TEA ONLY		COFFEE ONLY		NEITHER	
p T	q T	p T	q F	p F	q T	p F	q F
$\wedge$ T	$\vee$ T	$\wedge$ F	$\vee$ T	$\wedge$ F	$\vee$ T	$\wedge$ F	$\vee$ F
$\oplus$ F	neither F	$\oplus$ T	neither F	$\oplus$ T	neither F	$\oplus$ F	neither T

You just built a truth table

n atoms $\rightarrow 2^n$ worlds

For two atoms: TT • TF • FT • FF. No case is skipped.

TRUTH TABLE

One table exposes every connective's pattern

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	F	T	T	F
T	F	F	F	T	T
F	T	T	F	T	T
F	F	T	F	F	F

AND: TT only • OR: all but FF • XOR: TF and FT

EXERCISE 3

When do these operator claims hold?

Let p and q be propositions. Test all four input worlds.

$$p \oplus q \neq p \vee q$$

WHEN?

TT only

$$p \vee q = p \wedge q$$

WHEN?

TT and FF

$$p \oplus q = p \wedge q$$

WHEN?

FF only

EXERCISE 4

Keys, umbrellas, and emergency cookies

Translate each sentence. Do not evaluate it.

1

P: I find my keys • Q: I'm late for work

I'm not late for work and I found my keys.

$\neg Q \wedge P$

2

P: It's raining • Q: I forgot my umbrella

It's not raining or I forgot my umbrella.

$\neg P \vee Q$

3

P: I have cookies • Q: I have milk

I have cookies and milk.

$P \wedge Q$

EXERCISE 5

Build the truth table for this expression

$$(p \vee q) \wedge \neg(p \wedge q)$$

Compute one row at a time. Then compare your result column.

p	q	$p \vee q$	$p \wedge q$	$\neg(p \wedge q)$	result
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

SAME OUTPUT IN ALL FOUR WORLDS

Surprise: you just reinvented XOR

$$(p \vee q) \wedge \neg(p \wedge q) \equiv p \oplus q$$

Your result column matches $p \oplus q$ in every row.

DE MORGAN'S LAWS

Negation flips the gate and visits both inputs

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

PUSH Move $\neg$ through the parentheses.

VISIT BOTH Negate p and negate q .

FLIP $\wedge$ becomes $\vee$; $\vee$ becomes $\wedge$.

VERIFY The original and rewrite match in every row.

DE MORGAN • YOUR TURN

Push each NOT all the way inward

Rewrite each expression so $\neg$ applies only to p or q .

$$\neg(p \wedge \neg q)$$

$$\neg p \vee q$$

Flip $\wedge$ to $\vee$; double negation restores q .

$$\neg(\neg p \vee q)$$

$$p \wedge \neg q$$

Flip $\vee$ to $\wedge$; double negation restores p .

MOVE 04 • PROMISE

— Implication: if p , then q —

Only one world breaks it.

$p \rightarrow q$

A conditional is a promise about one case

Rain may or may not arrive. The promise matters only if it does.

The duck is not part of the proof. The duck has tenure.



EXERCISE 6

Which world breaks the promise?

p = submitted by 5

q = confirmation arrives

If submitted by 5, then confirmation arrives.

1	2	3	4
submit → confirmation	submit → no confirmation	no submit → confirmation	no submit → no confirmation

Only "submit + no confirmation" breaks the promise.

EXERCISE 7

Name each world: kept, broken, or unviolated

p: submitted by 5 • q: confirmation arrives

WORLD 1	WORLD 2	WORLD 3	WORLD 4
$p T \rightarrow q T$	$p T \rightarrow q F$	$p F \rightarrow q T$	$p F \rightarrow q F$
kept $p \rightarrow q = T$	broken $p \rightarrow q = F$	no violation $p \rightarrow q = T$	no violation $p \rightarrow q = T$

When p is false, the conditional is vacuously true.

IMPLICATION • $p \rightarrow q$

Only $T \rightarrow F$ makes a conditional false

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

A conditional proposition is false only when the premise is true and the conclusion is false.

TRIVIALY TRUE CONDITIONAL

A true conclusion makes a conditional trivially true

$$\mathbf{q = T \Rightarrow p \rightarrow q = T}$$

Once the conclusion is true, the premise cannot spoil the conditional.

If 91 is prime, every square has four sides.

F $\rightarrow$ T

The conclusion carries the truth.

If 7 is prime, every square has four sides.

T $\rightarrow$ T

The conclusion still carries the truth.

SAME TRUE CONCLUSION • WILDLY DIFFERENT PREMISES

Trivially true does not mean meaningfully related

If this laptop is purple, 11 is prime.

T

11 is prime regardless.

If every cat codes in Rust, 11 is prime.

T

11 remains stubbornly prime.

If the moon is cheese, 11 is prime.

T

Astronomy cannot unsettle arithmetic.

Truth tables track truth—not relevance, causation, or conversational good manners.

VACUOUSLY TRUE CONDITIONAL

A false premise makes a conditional vacuously true

$$p = F \Rightarrow p \rightarrow q = T$$

The promise is never activated, so the broken-promise row cannot occur.

If 9 is even, $2 + 2 = 5$.

The premise never starts the promise.

$F \rightarrow F$

If 9 is even, every square has four sides.

Still no violation.

$F \rightarrow T$

EMPTY CASES STILL OBEY THE RULE

Vacuous truth powers real proofs

Every dragon in this room submitted the worksheet.

T

There are no dragons here.

Every integer between 2 and 3 is even.

T

There are no such integers.

If a square has five sides, it has area 17.

T

No square enters the bad row.

No counterexample exists because no object satisfies the premise.

A conditional can be true for two easy reasons

If Lahore is in Pakistan, then $2 + 2 = 4$.

$$T \rightarrow T = T$$

q is true $\rightarrow$ trivially true

If 8 is odd, then $2 + 2 = 4$.

$$F \rightarrow T = T$$

Both easy routes apply

If 8 is odd, then $2 + 2 = 5$.

$$F \rightarrow F = T$$

p is false $\rightarrow$ vacuously true

True conclusion: trivial. False premise: vacuous. Neither is a causal explanation.

NEGATING A CONDITIONAL

NOT isolates the one broken-promise world

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$p \wedge \neg q$
T	T	T	F	F
T	F	F	T	T
F	T	T	F	F
F	F	T	F	F

To deny “if p, then q,” assert p and show that q failed.

FOLLOW THE GUARANTEE

"Only if" points to the requirement

A if B

$$B \rightarrow A$$

B is sufficient for A

A only if B

$$A \rightarrow B$$

B is necessary for A

You enter only if the QR code scans: enter $\rightarrow$ scan.

The promise, decoded

1 Propositions

A fixed declarative claim has exactly one truth value.

2 Connectives

NOT, AND, OR, and XOR turn truth values into precise rules.

3 Truth tables

Test every possible world instead of trusting a convenient example.

4 Conditionals

$p \rightarrow q$ fails only when p is true and q is false.

5 Easy truths

A true conclusion or a false premise makes a conditional true.

6 Only if

In “ p only if q ,” q is the requirement: $p \rightarrow q$.

Next: necessary and sufficient conditions, the conditional family, and equivalent forms.